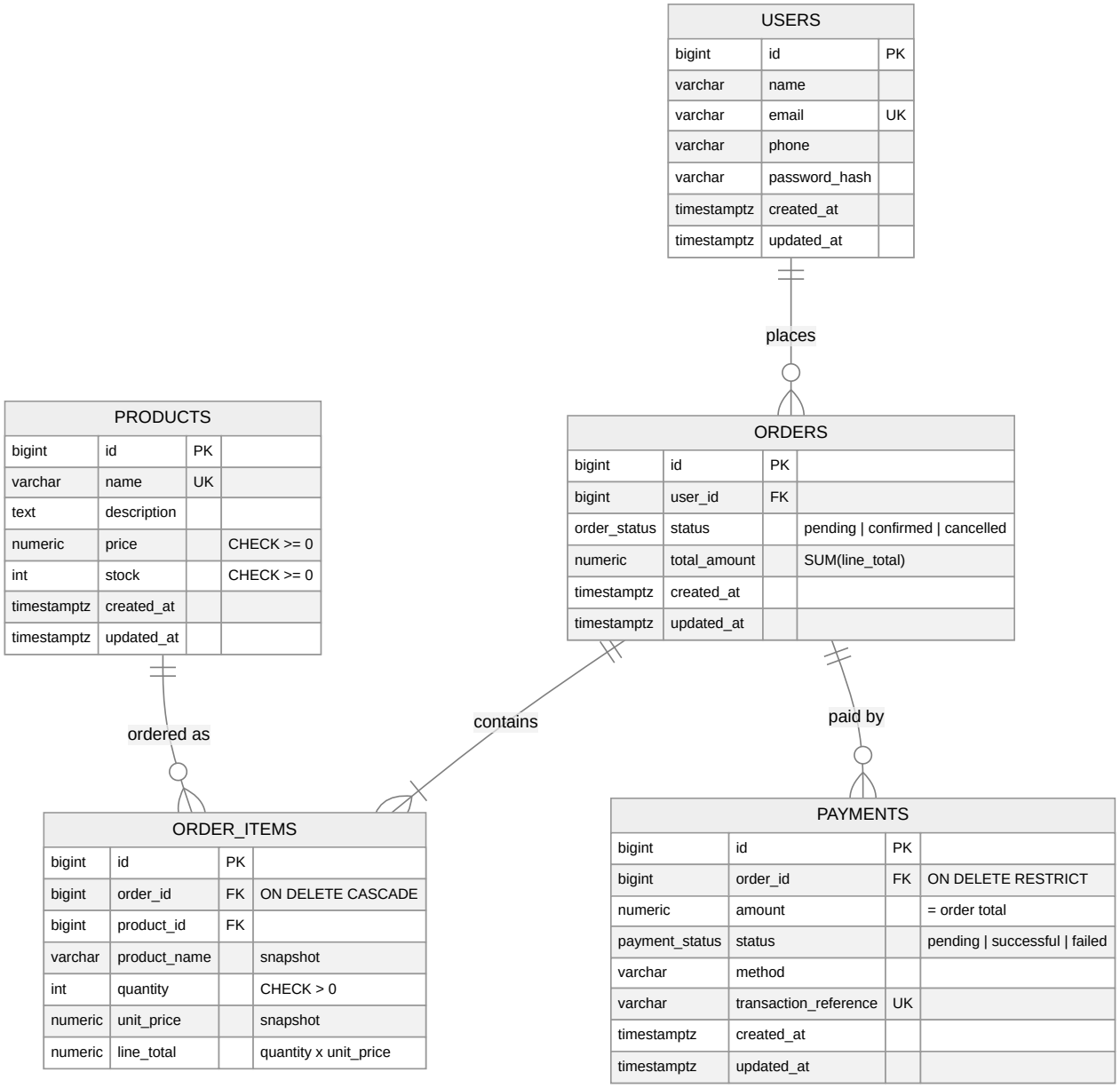
Orders & Payments: ER Diagram

PostgreSQL · PK = primary key, FK = foreign key, UK = unique



- **Order total:** each item's *line_total* = quantity × unit_price; a trigger keeps *orders.total_amount* = sum of line totals.
- **Pay only confirmed orders:** trigger on *payments* rejects a payment unless the order is *confirmed* and the amount equals the total.
- **Delete only orders without payments:** *payments.order_id* FK is ON DELETE RESTRICT; items are removed with the order (CASCADE).
- **Stock:** adding an item subtracts its quantity from *products.stock* (rejected if not enough); cancelling or deleting the order returns it.
- **One successful payment per order:** partial unique index on *payments(order_id) WHERE status = 'successful'*.